

Inverse Butterfly 1-2-1 (SPX)

Presentation report · Prop Desk Quant · 2026-06

QuantConnect backtest · 2021-2026 · SPX weeklys · hourly data (validation stage)

WHAT IT IS — a long-volatility structure (the inverse of an iron fly).

Per unit: +2 calls at-the-money / -1 call (ATM-W) / -1 call (ATM+W). Net CREDIT.

It MAKES money when the market MOVES (either direction) and LOSES if it sits still. Long gamma/vega. Width W set in multiples of the implied 1-sigma move.

WHAT WE TESTED — horizons 1/4/7/15/30/45 DTE; widths 0.15 / 0.25 / 0.40 sigma; exits by DTE-remaining, expiry-day clock, and profit-target. (German 'Castle Trader' income version.)

HEADLINE

1. Execution is CLEAN (verified): near-ATM legs are liquid, real spread ~\$150/trade. The scary -\$175k hourly 'worst case' is a stale-quote ARTIFACT (page 4) — opposite of PL5.
2. But the HOLD edge is THIN: mid hold is only +\$6-14k/5y and does NOT survive even light slippage; the daily 1/4-DTE variants accumulate cost over 1,200+ trades and go negative.
3. The VIABLE form is profit-target + WIDE wings: TP 50% is +\$21-74k; 0.40-sigma holds +\$38k. Structural headwind: implied vol > realized (0.73-0.85) — you overpay for movement.

Still in validation — NOT in the app yet.

1. Results by horizon (mid pricing)

net P&L over 5 years · 'realistic' = mid minus modeled real slippage (~\$150/trade, see p.4)

DTE	Trades	Hold (mid)	Hold (realistic)	TP 50%	Realiz/Impl
1 DTE	1239	\$+9,940	\$-82,985	\$+46,772	0.85
4 DTE	1290	\$+10,102	\$-86,648	\$+73,992	0.81
7 DTE	243	\$-500	\$-18,725	\$+20,964	0.75
15 DTE	254	\$+14,094	\$-4,956	\$+32,686	0.75
30 DTE	185	\$+5,714	\$-8,161	\$+25,330	0.77
45 DTE	178	\$+8,668	\$-4,682	\$+38,978	0.73

Hold at mid is modest and noisy; net of the verified ~\$150/trade spread it goes marginal/negative (the daily 1/4-DTE variants accumulate cost over 1,200+ entries). TP 50% is shown at MID and is GROSS of the exit spread (closing at the target also pays slippage) — net of the round-trip it is marginal-to-positive, best at wide width. Realiz/Impl < 1 = implied vol richer than realized.

2. Results by width (30 DTE, mid pricing)

wider wings = bigger profit per move, lower hit rate

Width	approx W	Win rate	Hold (mid)	Hold (realistic)	TP 50%
0.15 σ	~30 pts	90%	\$+5,714	\$-8,161	\$+25,330
0.25 σ	~50 pts	86%	\$+20,175	\$+6,600	\$+41,214
0.40 σ	~80 pts	79%	\$+38,138	\$+25,613	\$+43,394

Widening the wings raises the dollar payoff per move (+\$6k -> +\$20k -> +\$38k at mid) while the win rate falls (90% -> 79%). The wider structures survive the modeled slippage best, so width is the most promising structural lever to lift the edge above execution cost.

4. Execution check — we verified the spread (the PL5 lesson)

methodology — verify, don't assume

On PL5 we learned NOT to trust an aggregate, and NOT to assume. So we verified this one.

THE WORRY. At hourly data, the worst-case ('cross every leg') hold looks catastrophic:

30 DTE: +\$5.7k at mid vs -\$175k at full hourly bid/ask. A 4-leg structure shouldn't cost that.

THE TEST. We re-ran 30 DTE on MINUTE data (2025) and measured the entry spread per trade:

- MINUTE spread: consistent, ~\$115-310/trade (median ~\$150).
- HOURLY spread: erratic, \$95 up to \$5,580/trade (median ~\$1,090) — stale-quote spikes.
- Where the hourly quote happened to be tight, it matched minute; where stale, it inflated 7-37x.

THE VERDICT. These legs are near-ATM and LIQUID. The hourly 'cons' is a STALE-QUOTE ARTIFACT;

the real execution cost is small (~\$150/trade). So we IGNORE the -\$175k and use the modeled ~\$150/trade for the 'realistic' columns. (This is the OPPOSITE of PL5, whose -3-delta tail had a genuinely wide spread that held up at minute — there the worst case was real.)

Why hourly lies here: a far-from-busy option's last hourly quote can be stale/wide; near-ATM minute quotes refresh constantly. Same data vendor, different sampling — minute is the truth.

5. Verdict (preliminary — still validating)

honest conclusion

Execution is clean (verified), but the BUY-AND-HOLD edge is too thin to survive even light slippage. The strategy is only viable as PROFIT-TARGET + WIDE wings — not hold-to-expiry.

What we can say with the data:

- Execution is NOT the problem (verified): real spread ~\$150/trade, not the -\$175k artifact.
- The lever that works is WIDTH: 0.40 sigma holds +\$38k at mid, most robust after slippage.
- TP 50% beats hold at mid (+\$21-74k); net of round-trip slippage it is marginal-to-positive (best at wide width) — taking profit is still the right exit, but the net edge is borderline.
- The headwind is structural: Realiz/Impl 0.73-0.85 — you overpay for movement.

Honest caveats:

- Hourly data; the minute check validates the spread but full-span numbers remain hourly.
- Thin 30-DTE hold turns marginal after the (small, real) slippage — width/TP carry the edge.
- Long-vol P&L is lumpy (a few big moves make the year) — needs the lumpiness shown forward.

Suggested next step: forward-test the WIDE (0.40 sigma) variant with a profit-target exit, on a few liquid index underlyings — execution is clean, so the open question is purely the edge.